

MERAKER-EGGE, Norway

WMO: 012930

Lat: 63.411N

Lon: 11.727E

Elev: 561

StdP: 14.40

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	0.1	-10.3	3.2	-5.5	-4.8	4.3	1.2	23.0	33.0	20.5	31.8	3.8	190	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	79.9	62.0	75.9	60.4	72.1	58.6	64.1	76.0	62.0	72.5	60.0	69.7	6.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.2	76.8	68.9	57.3	71.5	66.0	55.5	67.0	63.7	29.5	75.8	28.0	72.7	26.6	69.9	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.3	13.0	-9.7	85.3	5.1	4.2	-13.4	88.3	-16.3	90.7	-19.2	93.1	-22.9	96.1		
(5)			-10.1	67.6	5.0	2.5	-13.7	69.4	-16.7	70.9	-19.5	72.3	-23.2	74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	24.9	26.2	30.1	38.3	46.5	53.2	59.2	56.7	50.2	40.5	31.7	27.1		
	DBStd	14.44	10.86	10.44	8.93	5.70	7.08	6.21	6.24	5.27	5.51	6.92	9.44	10.93		
	HDD50	4238	779	667	617	351	157	34	2	6	63	305	548	709		
	HDD65	8982	1244	1087	1082	800	573	358	200	262	443	761	998	1174		
	CDD50	759	0	0	0	1	49	129	288	213	70	9	0	0		
	CDD65	27	0	0	0	0	0	2	21	4	0	0	0	0		
	CDH74	544	0	0	0	0	35	87	345	76	1	0	0	0		
	CDH80	104	0	0	0	0	3	13	82	6	0	0	0	0		
	WSAvg	5.3	6.0	5.9	5.9	5.3	5.3	5.2	4.7	4.6	5.0	5.2	5.1	5.6		
	PrecAvg	40.1	3.4	2.6	2.8	2.0	2.5	3.5	4.5	3.8	4.8	3.7	3.1	3.8		
(15) Precipitation	PrecMax	48.2	9.5	4.8	9.4	4.6	4.8	5.9	7.6	6.3	9.1	5.5	7.6	8.7		
	PrecMin	32.0	0.0	0.1	0.5	0.6	0.4	1.1	2.4	1.4	1.5	1.7	0.0	1.3		
	PrecStd	4.5	2.2	1.4	2.2	1.1	0.9	1.4	1.4	1.4	1.9	1.0	2.0	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	46.7	51.6	63.4	79.1	82.3	87.6	80.8	72.1	62.7	54.3	47.4		
		MCWB	40.1	40.0	41.3	49.4	59.3	60.8	66.6	62.2	57.5	53.3	48.0	41.8		
	2%	DB	42.3	43.3	46.4	58.2	72.2	75.8	81.6	75.8	68.0	57.8	50.0	44.9		
		MCWB	37.9	37.9	39.7	46.0	55.8	58.6	63.7	60.7	57.4	49.7	45.0	40.1		
	5%	DB	40.3	41.1	43.9	53.7	66.8	71.1	77.7	72.1	64.3	54.9	47.0	42.6		
		MCWB	36.7	36.7	38.7	43.3	53.9	57.3	61.9	59.6	55.4	48.5	42.5	38.3		
	10%	DB	38.0	39.0	41.3	49.2	60.8	66.7	73.6	68.4	60.6	51.4	44.0	40.4		
		MCWB	34.8	35.3	36.5	41.3	50.3	55.1	60.4	58.2	53.1	46.5	39.6	36.6		
	0.4%	WB	41.1	41.1	43.3	50.1	60.3	63.0	68.8	64.9	60.3	55.2	48.7	43.0		
		MCDB	43.7	44.8	47.3	62.2	75.9	75.9	83.7	75.7	68.8	58.6	53.0	45.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	39.0	41.3	46.9	57.1	60.4	65.8	62.7	58.1	50.7	45.7	40.8		
		MCDB	41.4	42.5	45.6	56.4	69.4	72.8	79.3	72.5	66.3	56.5	49.4	44.2		
	5%	WB	37.1	37.3	39.1	44.3	54.1	58.2	63.2	60.8	56.0	48.8	42.8	38.8		
		MCDB	39.7	40.0	43.3	51.9	65.3	68.7	74.7	69.4	62.6	54.1	46.6	41.9		
	10%	WB	34.9	35.4	37.1	42.1	51.0	56.0	61.1	59.0	53.8	46.8	39.9	36.7		
		MCDB	37.5	38.8	40.9	48.3	60.5	65.1	71.4	66.7	59.9	51.4	43.4	39.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.1	15.6	17.3	18.3	17.5	19.2	18.9	16.8	13.3	11.1	11.3		
		MCDBR	10.2	10.6	16.5	25.7	30.0	27.7	29.3	26.3	23.8	17.7	14.7	12.6		
		MCWBR	7.6	7.5	10.9	15.3	15.7	13.7	14.2	13.9	14.3	11.4	10.6	9.9		
		MCDBR	9.8	10.4	14.6	21.9	27.9	24.5	26.8	22.6	22.4	15.7	14.5	12.4		
	5% WB	MCWBR	7.8	8.1	10.2	13.5	15.1	12.7	13.7	12.7	13.9	10.6	10.8	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.219	0.280	0.313	0.342	0.351	0.349	0.358	0.346	0.321	0.300	0.230	N/A		
	taud		2.007	2.179	2.201	2.257	2.377	2.450	2.444	2.473	2.529	2.418	2.083	N/A		
	Ebn at Noon		124	205	246	265	273	277	270	264	249	203	118	N/A		
	Edh at Noon		10	19	29	34	33	31	31	27	21	16	9	N/A		
(44) All-Sky Solar Radiation	RadAvg		51	220	627	1198	1454	1519	1473	1127	675	297	80	24		
	RadStd		7	33	84	95	99	151	155	126	68	38	12	2		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A	N/A

Nomenclature: See separate page